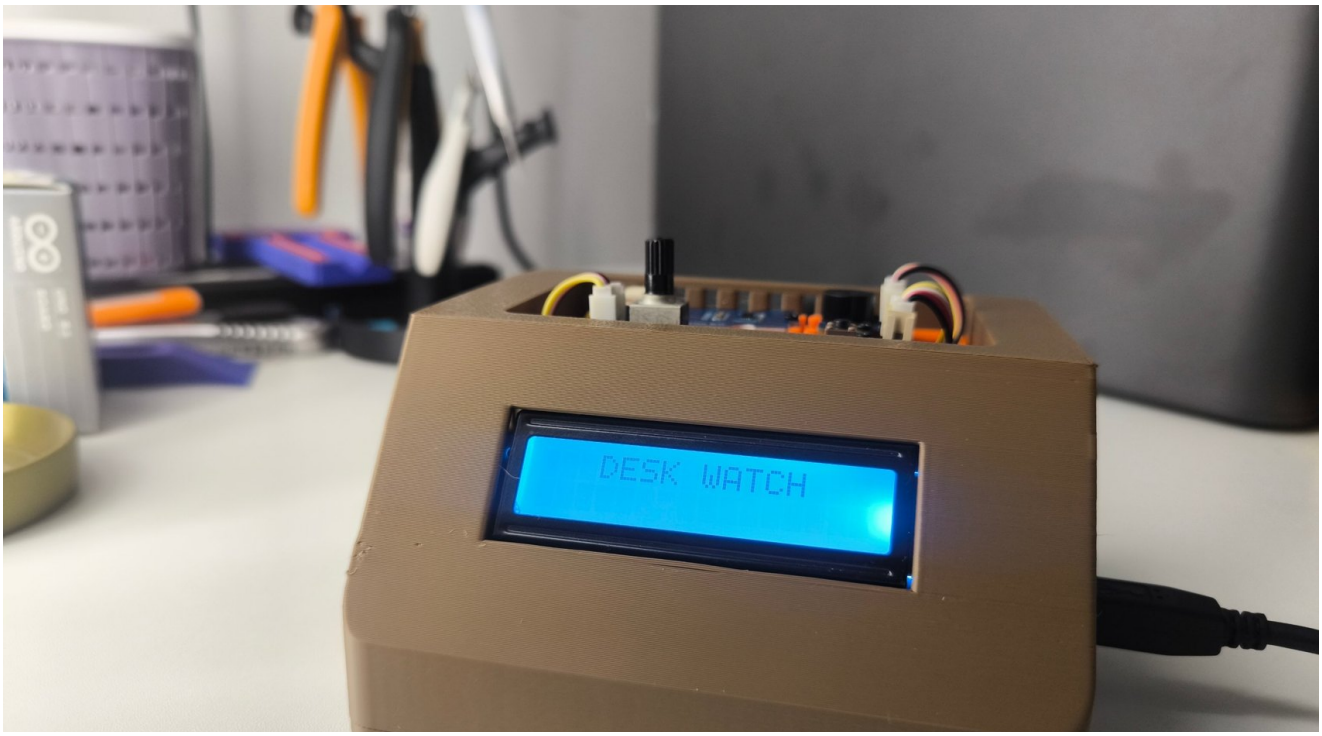
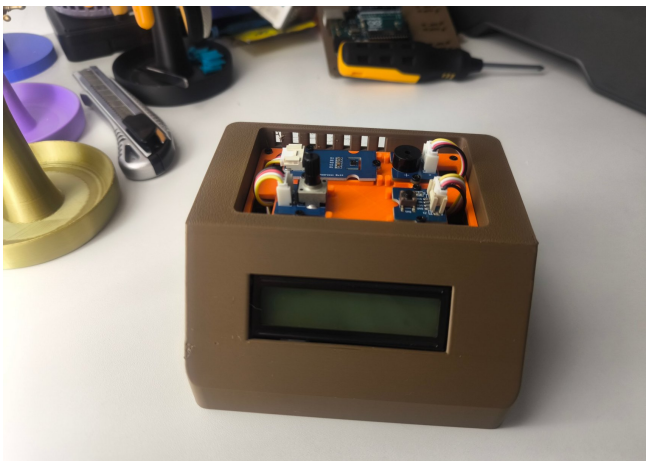
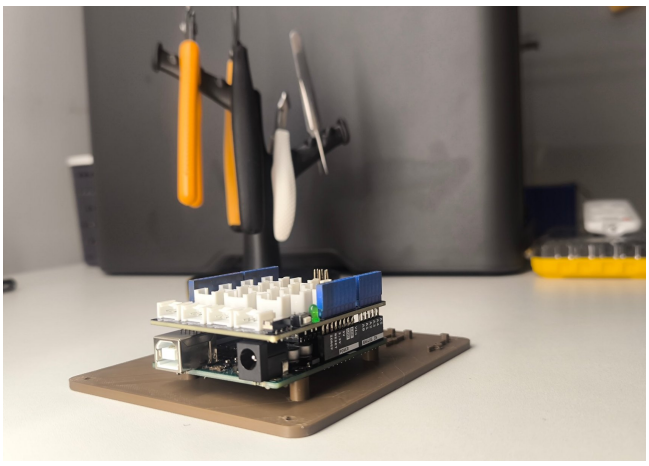
Desk Watch 1

User guide and build reference

Arduino UNO + Grove desk console | Pr. BELAMINE MakerLab

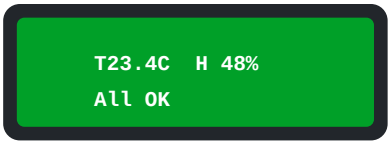




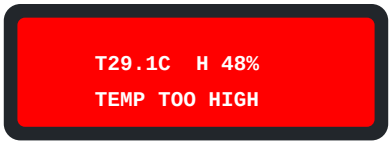


Printed prototype: start-up screen, open top plate with the four Grove modules, and the Arduino UNO with its Base Shield on the base.

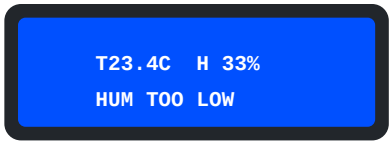
Desk Watch 1 is a compact desk console built around an Arduino UNO and a Grove Base Shield. It shows temperature and humidity on a 16x2 colour LCD, warns you when the room goes out of range, and doubles as a timer, stopwatch, hand counter, dice and chess clock. Everything is driven by **one button and one dial**, and every module plugs in with a Grove cable: **no soldering**.



Everything in range



Temperature too high



Humidity too low

At a glance

- **Climate monitor:** SHT31 sensor, adjustable HIGH / LOW thresholds, colour and buzzer alerts.
- **Tools:** Pomodoro timer, stopwatch, hand counter, dice (D2 to D100), two-player chess clock.
- **Clock and alarm** with the optional RTC module; night mode and automatic sleep.
- **4 languages** (English by default, French, Spanish, German) and a **27-page manual built into the firmware**.
- **Fits an Arduino UNO:** the sketch uses about 93 % of the program memory and 22 % of the RAM, with no extra library besides the Grove LCD one.

Status of this release: the sketch compiles for the Arduino UNO and boots on the prototype; fine tuning on the real device (buzzer loudness, dial feel) is still in progress. Enclosure files are listed on the model page.

1. What you need

Part	Notes
Arduino UNO R3	Any UNO R3 compatible board
Grove Base Shield for Arduino	Keep the voltage switch on **5 V**
Grove LCD RGB Backlight V4.0 (16x2)	I2C. Uses the *Grove - LCD RGB Backlight* library (rgb_lcd.h)
Grove SHT31 temperature and humidity sensor	I2C, address 0x44
Grove Buzzer	Digital port D3
Grove Button	Digital port D4
Grove Rotary Angle Sensor (the dial)	Analog port A1
Grove RTC (DS1307 or DS3231), optional	I2C. Needed for the clock, the alarm and the automatic night mode
Grove cables, USB-B cable, small screws	See the assembly guide for the screw plan

2. Wiring

Plug each Grove cable into the matching port of the Base Shield. The three I2C devices (LCD, SHT31 and the optional RTC) can share any I2C ports: they are wired in parallel.

Module	Base Shield port	Signal
LCD RGB Backlight	I2C	SDA / SCL
SHT31	I2C	SDA / SCL (address 0x44)
RTC (optional)	I2C	SDA / SCL (address 0x68)
Buzzer	D3	Digital output, software square wave
Button	D4	Digital input, HIGH when pressed
Rotary Angle Sensor	A1	Analog input, 0 to 1023

The small **L LED on the Arduino board (pin 13)** lights up while the Desk Watch is in standby, and while you hold the button long enough to put it to sleep. Its colour depends on the board (often orange). To get a red indicator, plug a red Grove LED on a free port and change `PIN_LED` at the top of the sketch.

3. Installing the firmware

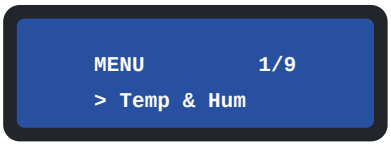
Step	What to do
1	Install the Arduino IDE (2.x), then in the Library Manager install **Grove - LCD RGB Backlight** (Seeed Studio). No other library is needed: the SHT31 and the RTC are read directly over I2C.
2	Open `Desk_Watch_1.ino`, choose **Arduino UNO** and your port, then upload.
3	At power-up you see DESK WATCH, a short melody, then PR BELAMINE MAKERLAB and a module check (RTC and sensor).
4	If the dial turns the wrong way for your wiring, invert it in `potRead()`; if your dial is on another port, change `PIN_POT`.

4. Controls

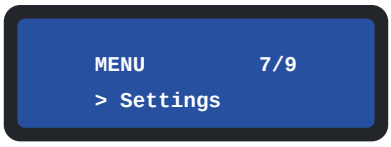
The Desk Watch has one button and one dial. The same gestures work on every page.

Gesture	What it does
Short press	Open the menu from the home screen; select the entry in the menu; confirm a setting. A short press is confirmed 0.3 s after release (the double-click delay).
Double click	Page action: reset the timer or stopwatch, set the thresholds on Temp & Hum, arm or disarm the alarm, pause the chess clock. Inside Settings it goes back to the menu.
Long press (0.8 to 2.5 s)	Back to the menu from any page. Timers and the stopwatch keep running in the background.
Very long press (over 2.5 s)	Standby: the screen switches off and the L LED lights up. Any button press wakes it. The alarm stays active.
Dial	In the menu, scrolls the entries. On a page, chooses a value (timer length, dice sides, chess time control, thresholds...). Turning right moves to the next entry or a larger value.

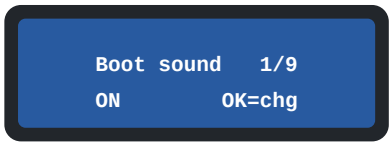
The menu



Menu, first entry



Menu, Settings



Settings list

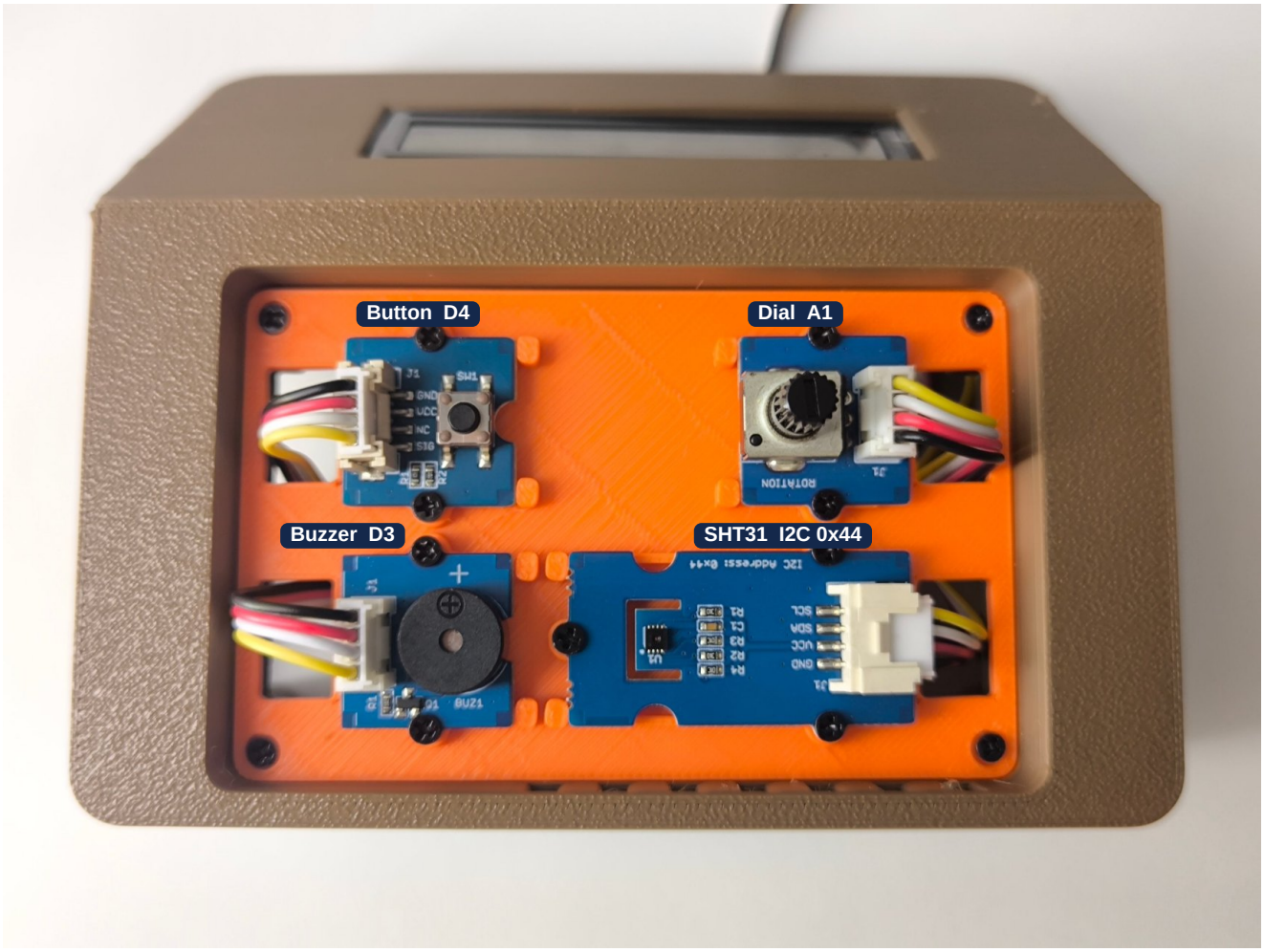
#	Entry	What it opens
1	Temp & Hum	Temperature, humidity and alerts
2	Timer	Pomodoro timer, 1 min to 4 h
3	Stopwatch	Chronograph
4	Counter	Hand tally counter
5	Dice	Dice from 2 to 100 sides
6	Chess	Two-player chess clock
7	Settings	Sound, language, home page, night mode, sleep, alarm, clock, brightness
8	Manual	27-page built-in manual, scroll with the dial
9	Home	The page you chose in Settings (automatic by default)

Top-right badges

When a function runs in the background, a letter appears in the top-right corner of the screen: **!** a temperature or humidity alert, **A** alarm armed, **C** stopwatch running, **P** timer running (the P blinks for 1 s every 10 s).

Module layout

Top plate seen from above, the screen being at the top of the picture. Each module is held by two screws and plugs into the Base Shield with a short Grove cable that runs down through the side slots.

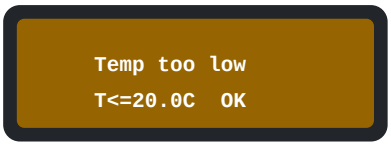


5. The pages

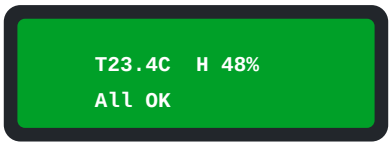
Temp & Hum

The home page of the station. Colours tell you the state at a glance: **green** all in range, **red** too high, **blue** too low. The buzzer gives two short beeps for HIGH and one long beep for LOW (silent in night mode and in standby). A small hysteresis (0.3 C and 2 %) stops the alert from flickering around a threshold.

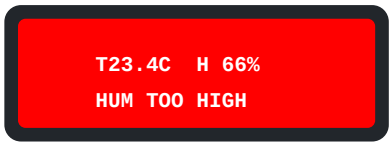
- **Double click** to set the four thresholds with the dial: temperature low and high (10 to 40 C, steps of 0.5 C), humidity low and high (10 to 90 %). A short press moves to the next one, the last one saves.
- Defaults: 20.0 to 28.0 C and 40 to 60 % humidity. Thresholds are stored and reused at start-up.



Threshold edit



In range



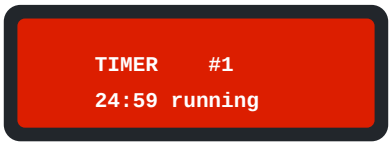
Alert

Timer (Pomodoro)

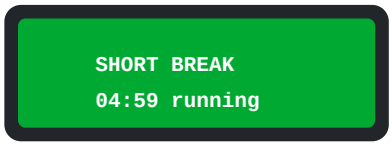
Turn the dial to choose the work time: 1 to 60 minutes in 1-minute steps, then 65 minutes to 4 hours in 5-minute steps. Short press starts or pauses; double click resets. After each work period comes a 5-minute break (a 15-minute break every 4th time). If you turn the dial or press long, the timer keeps running in the background.



Choosing the length



Running



Break

Stopwatch and Counter

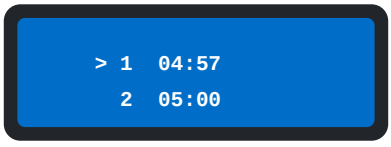
- **Stopwatch:** short press starts or stops, double click resets. Already at zero, a double click or a turn of the dial goes back to the menu.
- **Counter:** every press adds 1 immediately (up to 9999), even when you click fast. To reset, turn the dial: the screen asks to confirm, a short press means yes, a double click or 5 s without action means no.

Dice and Chess clock

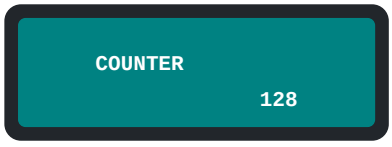
- **Dice:** the dial chooses the number of sides (2 for heads or tails, 4, 6, 8, 10, 12, 20, 30 to pick a student in a class of 30, or 100). A press rolls: the digits scroll with a small tick for about a second, then the result stays with a beep.
- **Chess clock:** the dial chooses the time control (1+0, 3+0, 3+2, 5+0, 5+3, 10+0, 10+5, 15+10, 30+0, 60+0: minutes + seconds added per move). A short press starts with player 1; each press then hands over to the other player and adds the increment. Double click pauses; when a flag falls the screen turns red with three beeps. Leaving the page pauses the clock, and the automatic sleep never triggers during a game.



Dice



Chess, player 1



Counter

6. Settings

Open **Settings** in the menu. Turn the dial to choose a line, short press to open it, turn the dial to choose the value, short press to confirm. A long press or a double click cancels the change and goes back.

#	Setting	Choices	Notes
1	Boot sound	On / Off	Plays a three-note melody at start-up and wake-up
2	Volume	Low / Medium / High	Changes the buzzer duty cycle; a test beep plays while you choose
3	Language	English, Francais, Espanol, Deutsch	English by default; applies to all screens and the manual
4	Home	Auto, Temp & Hum, Clock, Timer, Stopwatch, Counter	Page opened by Home in the menu and at start-up. Auto alternates the clock (5 s) and Temp & Hum (4 s) when an RTC is present; otherwise Temp & Hum only
5	Night mode	Off / Auto 22-6 h / On	Screen dimmed to 8 %, no alert beeps, no start-up melody. The alarm still rings at full brightness. Auto needs the RTC
6	Auto sleep	Never, 1, 2, 5, 10, 15, 30 min, 1 h	Screen off after no press and no dial movement. Never by default
7	Alarm	Time and on/off	Needs the RTC. Double click on the alarm page arms or disarms it; a press while it rings snoozes 5 minutes, a long press stops it
8	Set time	Hours, then minutes	Writes the RTC
9	Brightness	8 levels, 5 to 100 %	Applies to the backlight colour of every page

Without the RTC

The station, timers, stopwatch, counter, dice and chess clock all work without the RTC. The clock, the alarm and the automatic night mode need it: the screen then says *Install RTC module for clock*.

7. Troubleshooting

Symptom	Check
Backlight on but no text at start-up	Re-install the Grove LCD library and re-upload; check the LCD cable is on an I2C port.
Sensor missing	SHT31 cable on an I2C port; the sensor must answer at address 0x44.
No RTC clock	The RTC module is optional; plug it into an I2C port and restart.
The dial goes the wrong way or jumps	Confirm the dial is on A1 and the shield switch is on 5 V. Invert it in potRead() if needed.
The buzzer is too quiet or too loud	Settings, Volume. Piezo buzzers differ; an active buzzer module ignores the duty cycle.
The screen is almost black at night	Night mode is on (Settings 5). Set it to Off, or turn the dial / press the button for the menu.
Compile error: not enough memory	The sketch needs about 93 % of the UNO flash. Do not add large features without removing something first.

Desk Watch 1 is part of the Pr. BELAMINE MakerLab range, made by a teacher as a base for classroom electronics projects. Feedback and ideas are welcome in the comments of the model page.